



Incident handler's journal

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this journal as a way to log the key takeaways about the different cybersecurity tools or concepts you encounter in this course.

Date: 21/1/2025	Entry: #1
Description	Documenting a Cybersecurity incident where a small US healthcare clinic experienced severe disruption of their business operations.
Tool(s) used	None
The 5 W's	Capture the 5 W's of an incident. • Who : Group of Unethical Hackers • What : Phishing emails containing ransomware were sent to employees, and organizational data is encrypted by hackers. • When : Tuesday, 9:00 am • Where : A healthcare clinic • Why : Employees clicked the phishing mail which allowed the malware to be installed to the computers, that included a ransom note , In which the unethical hackers encrypted the organization's files and demanded money in exchange for the encryption key.
Additional notes	To prevent future incidents, it is encouraged that all employees are educated with simple security measures to prevent cyber attacks.

Date: 31/1/2025	Entry: #2
Description	Analyzing a suspicious file that was deducted to be malware.
Tool(s) used	VirusTotal
The 5 W's	Capture the 5 W's of an incident. • Who : A malicious hacker • What : A file that contains malware that retrieved access to sensitive information in the company. • When : 1:15 p.m • Where : an employee's computer at a financial services company • Why : An employee entered the password to open a file in an email, When the employee opened the file, a malicious payload was then executed on their computer.
Additional notes	According to VirusTotal, 59/72 security vendors labeled the suspicious file as a malware threat.

Date: 1/2/2025	Entry: #3
Description	Documenting an incident involving data exfiltration in a retail company.
Tool(s) used	-

The 5 W's	Capture the 5 W's of an incident. • Who : A malicious hacker • What : The hacker gained unauthorized access to a company's customer PII and financial information through a vulnerability in their web application. The hacker then requested money and cryptocurrency in exchange for not releasing the data publicly. • When : 3:13 p.m., PT, on December 22, 2022 • Where : A retail company • Why : A vulnerability was discovered in the e-commerce web application. This vulnerability allowed the attacker to perform a forced browsing attack and access customer transaction data by modifying the order number included in the URL string of a purchase confirmation page. This led to unauthorized access to customer data.
Additional notes	The following are recommendations to prevent future attacks: • Regularly perform vulnerability scanning and penetration testing. • Implement and configure access control mechanisms correctly

Date: 2/2/2025	Entry: #4
Description	Overview of Splunk
Tool(s) used	Splunk is a powerful log analysis, monitoring, and security platform designed to collect, index, and analyze machine-generated data from various sources in real-time. Here are the use cases:

	Data Ingestion & Indexing • Collects data from various sources: servers, applications, network devices, security logs, IoT devices, etc. • Uses a schema-on-read approach, meaning data is indexed as raw text and structured when queried. Search & Analysis • Uses the Splunk Search Processing Language (SPL) to query and analyze data. • Allows real-time and historical searches for troubleshooting and insights. Visualization & Reporting • Dashboards, charts, and graphs for better data representation. • Automated reports and alerts. Security & Compliance • Splunk Enterprise Security (ES) helps in threat detection, incident response, and compliance management. • Supports SIEM (Security Information and Event Management) capabilities.
The 5 W's	• Who: N/A • What: N/A • Where: N/A • When: N/A • Why: N/A

Additional notes	Splunk has its own querying language called Search Processing Language (SPL). SPL is used to search and retrieve events from indexes using Splunk's Search & Reporting app. An SPL search can contain many different commands and arguments. For example, you can use commands to transform your search results into a chart format or filter results for specific information.
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Date: 2/2/2025	Entry: #5
Description	Overview of Google's SIEM solution, Chronicle
Tool(s) used	Google Chronicle is a cloud-native security analytics platform built on Google's infrastructure. It is part of Google Cloud's Security Operations (SecOps) suite and serves as a SIEM tool, helping organizations detect, investigate, and respond to security threats efficiently. Here are the key features: Massive Data Ingestion & Retention: Ingests petabytes of security telemetry (logs, network traffic, endpoint data) in real-time. Offers long-term data retention (up to 1 year) for historical threat hunting. Ultra-Fast Search & Analysis: Uses Google's BigQuery and machine learning to enable instant searches

	across massive datasets. No need to worry about indexing or storage limits like traditional SIEMs. 3. Automated Threat Detection (YARA-L): Uses YARA-L, a query language similar to YARA, for writing security detection rules. Automates correlation of events, anomaly detection, and threat intelligence integration. 4. Integrated Threat Intelligence: Leverages Google's VirusTotal, Mandiant Threat Intelligence, and third-party feeds to enrich logs. Automatically correlates Indicators of Compromise (IoCs) with security events. 5. Behavioral Analysis & Threat Hunting: Uses machine learning and entity analytics to detect suspicious patterns across endpoints, networks, and cloud environments. Provides Unified Timelines for deep forensic investigation. 6. Cloud-Native & Scalable: Runs on Google Cloud infrastructure, ensuring high availability and scalability. No need for on-premises hardware or storage management.
The 5 W's	• Who: N/A • What: N/A • Where: N/A • When: N/A • Why: N/A
Additional notes	There are two types of searches you can perform to find events in Chronicle, a Unified Data Model (UDM) Search or a Raw Log Search.

	You can perform a UDM search by typing your search, clicking on “Search,” and selecting “UDM Search.” Through a UDM Search, Chronicle searches security data that has been ingested, parsed, and normalized. If you can't find the information you are searching for through the normalized data, using a Raw Log Search will search through the raw, unparsed logs. You can perform a Raw Log Search by typing your search, clicking on “Search,” and selecting “Raw Log Search.” Because it is searching through raw logs, it takes longer than a structured search.
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Date: Record the date of the journal entry.	Entry: Record the journal entry number.
Description	Provide a brief description about the journal entry.
Tool(s) used	List any cybersecurity tools that were used.
The 5 W's	Capture the 5 W's of an incident. • Who caused the incident? • What happened? • When did the incident occur? • Where did the incident happen? • Why did the incident happen?
Additional notes	Include any additional thoughts, questions, or findings.